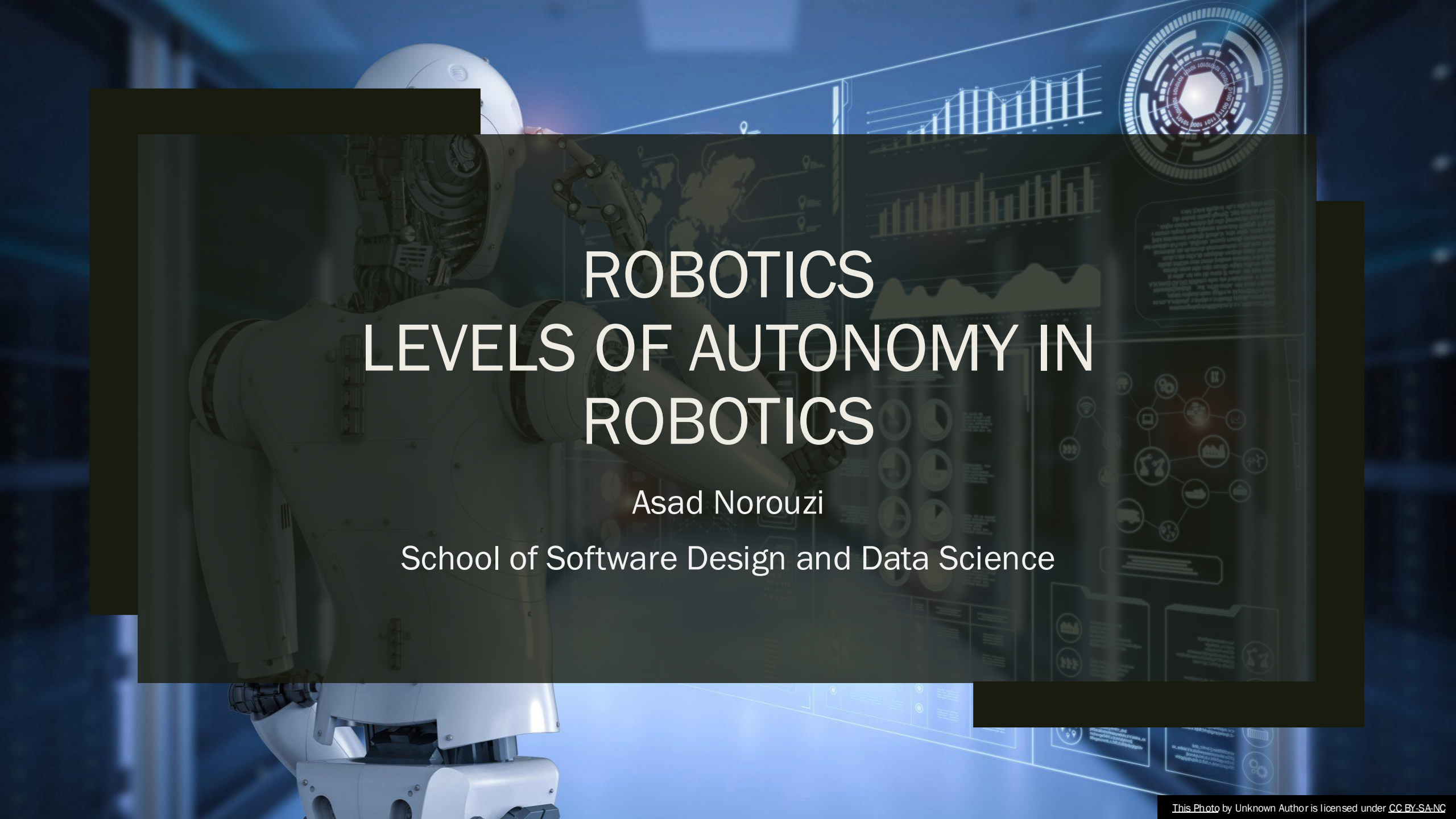


A white humanoid robot is shown from the waist up, positioned on the left side of the frame. It has a sleek, modern design with visible joints and a helmet-like head. The robot's right arm is raised, with its hand near its head. The background is a dark blue, futuristic environment with glowing digital elements. On the right side, there are several data visualizations: a bar chart at the top, a line graph below it, and a circular radar chart at the top right. A world map is also visible in the background. The overall atmosphere is high-tech and futuristic.

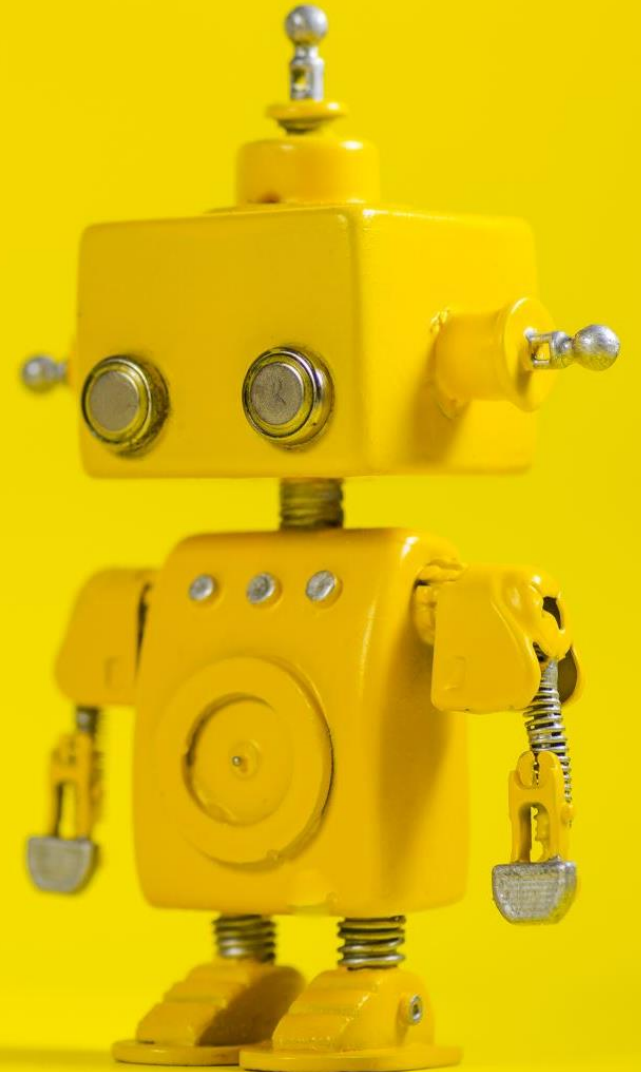
ROBOTICS LEVELS OF AUTONOMY IN ROBOTICS

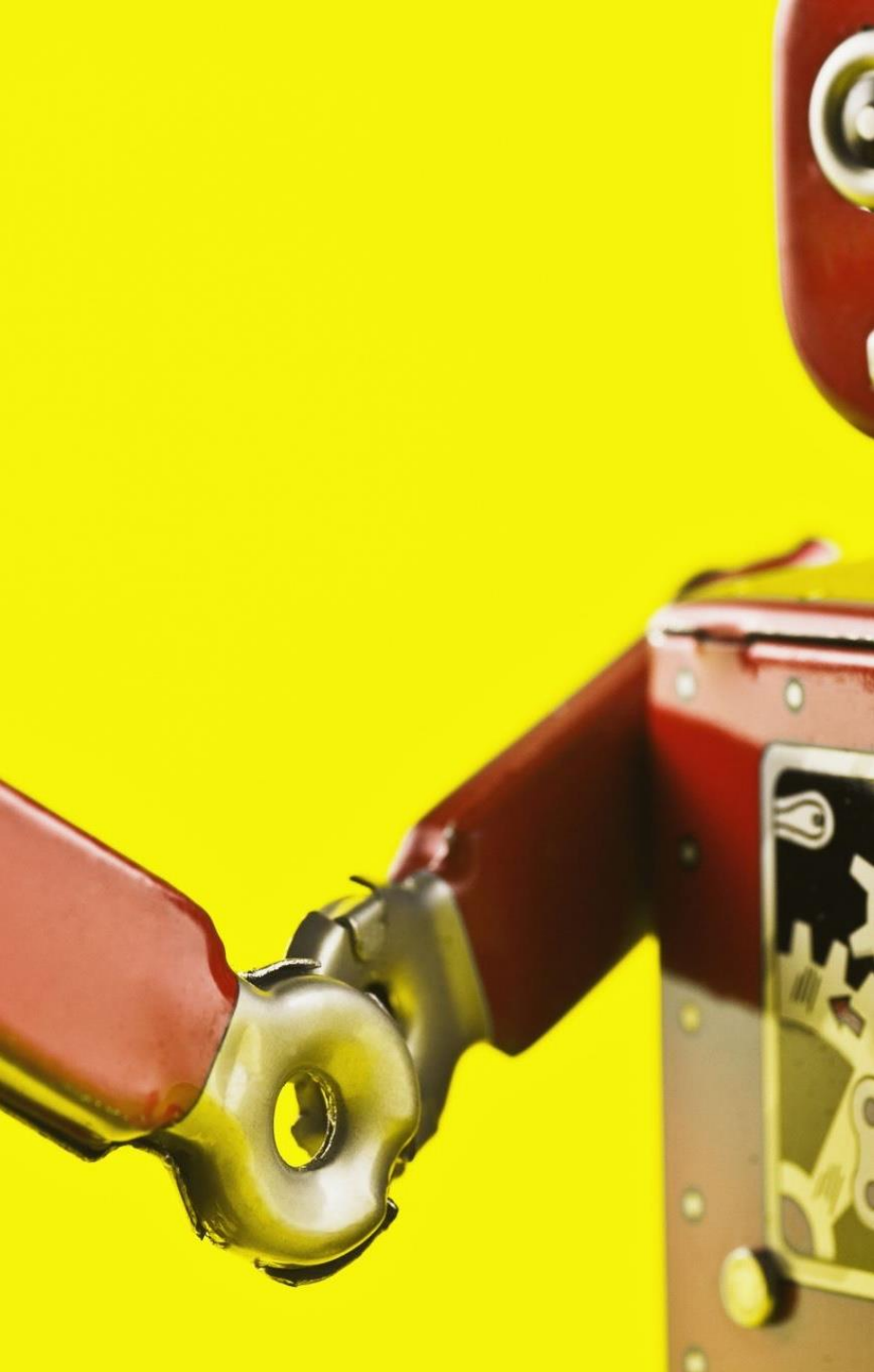
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Agenda

- Introduction to Autonomy in Robotics
- Levels of Autonomy Overview
- Level 0 – No Autonomy (Manual Control)
- Level 1 – Assisted Autonomy
- Level 2 – Partial Autonomy
- Level 3 – Conditional Autonomy
- Level 4 – High Autonomy
- Level 5 – Full Autonomy
- Applications and Implications of Autonomy Levels
- Summary
- Q&A





Introduction to Autonomy in Robotics

- Autonomy in robotics refers to the ability of a robot to perform tasks and make decisions without human intervention.
- Understanding the levels of autonomy helps in designing and implementing robotic systems suitable for different applications and environments.

Levels of Autonomy Overview

The levels of autonomy range from fully manual control to complete autonomy, where the robot operates independently of human intervention.

Various organizations, including SAE (Society of Automotive Engineers) and the U.S. Department of Defense, have defined levels of autonomy.

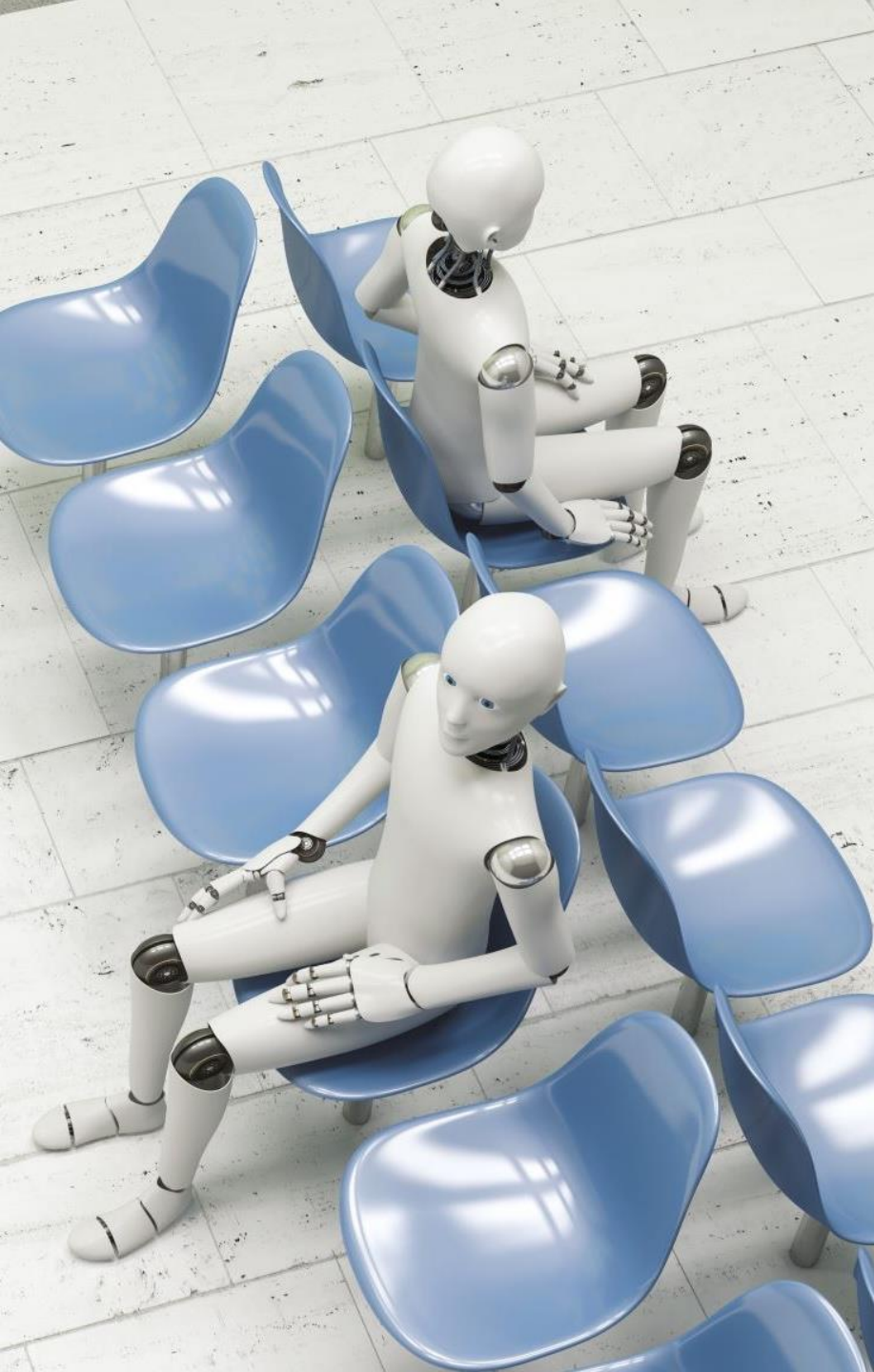
**Better
technology
makes more
better jobs for
horses**





Level 0 – No Autonomy (Manual Control)

- Robots are entirely controlled by human operators, with no autonomous capabilities.
- Examples:
 - *Teleoperated Robots: Remote-controlled drones or bomb disposal robots.*
 - *Industrial Manipulators: Robotic arms operated manually by a human worker.*
- Applications: Situations requiring high precision and human judgment, such as surgical robots and hazardous material handling.



Level 1 – Assisted Autonomy

- Robots perform basic tasks autonomously but require human intervention for complex decisions and actions.
- Examples:
 - *Robotic Vacuum Cleaners: Simple navigation and cleaning routines but require human setup and intervention for complex tasks.*
 - *Driver Assistance Systems: Features like cruise control and lane-keeping assistance in cars.*
- Applications: Household robotics, basic industrial automation, and driver assistance technologies.



Level 2 – Partial Autonomy

- Robots can perform a wider range of tasks autonomously but still need human oversight and intervention for critical decisions.
- Examples:
 - *Advanced Driver Assistance Systems (ADAS): Automated braking and adaptive cruise control systems in vehicles.*
 - *Semi-Autonomous Drones: Can follow predefined flight paths but require human control for takeoff, landing, and emergency maneuvers.*
- Applications: Automotive industry, aerial surveillance, and delivery drones.

Level 3 – Conditional Autonomy

- Robots can perform tasks autonomously under specific conditions but require human intervention in complex or unpredictable scenarios.
- Examples:
 - *Autonomous Vehicles (Level 3): Can drive themselves in controlled environments but need a human driver to take over in challenging conditions.*
 - *Robotic Surgery Systems: Perform surgical procedures with high precision but require a human surgeon to oversee and intervene if necessary.*
- Applications: Autonomous driving, medical robotics, and smart manufacturing systems.



Level 4 – High Autonomy

- Robots operate autonomously in most scenarios, with human intervention needed only in rare and complex situations.
- Examples:
 - *Fully Autonomous Vehicles (Level 4): Can navigate and drive in most environments without human intervention.*
 - *Warehouse Robots: Perform tasks like inventory management and order fulfillment with minimal human oversight.*
- Applications: Logistics, autonomous transportation, and advanced industrial automation.



Level 5 – Full Autonomy

- Robots operate completely autonomously, handling all tasks and decision-making processes without any human intervention.
- Examples:
 - *Fully Autonomous Vehicles (Level 5): Can operate in any environment and condition without human input.*
 - *Autonomous Exploration Robots: Operate independently in unstructured and unpredictable environments like space or deep-sea exploration.*
- Applications: Fully autonomous vehicles, space exploration, and advanced robotics in unstructured environments.

Applications and Implications of Autonomy Levels

Impact on Industries:

- Manufacturing: Improved efficiency and safety through automation of complex tasks.
- Healthcare: Enhanced precision and reduced human error in surgical procedures.
- Transportation: Increased safety and convenience through autonomous vehicles.

Ethical and Social Implications:

- Job Displacement: Potential impact on employment due to automation.
- Safety and Reliability: Ensuring the safe and reliable operation of autonomous systems.
- Regulation and Standards: The need for robust regulatory frameworks to manage the deployment of autonomous systems.

Summary

- **Levels of Autonomy:** From manual control (Level 0) to full autonomy (Level 5).
- **Applications:** Diverse applications in industries such as manufacturing, healthcare, transportation, and exploration.
- **Challenges:** Technical, ethical, and regulatory challenges.
- **Key Takeaways:**
 - *Understanding Autonomy: Importance of knowing the levels of autonomy in designing and deploying robotic systems.*
 - *Impact on Society: The potential benefits and challenges of autonomous systems.*



Q&A